

8. $(e^x+1)y' + e^x y = e^x - 1$ poišči reš., ki gre skozi izhodišče

Homogeni del:

$$(e^x+1)y' + e^x y = 0$$

$$\frac{y'}{y} = -\frac{e^x}{e^x+1}$$

$$\ln \frac{y}{c} = -\int \frac{e^x dx}{e^x+1} = -\int \frac{dt}{t} = -\ln(e^x+1)$$

$$\frac{y}{c} = \frac{1}{e^x+1} \rightarrow y_H = \frac{c}{e^x+1}$$

Variacija konstante: spl. reš. v obliki $y = \frac{c(x)}{e^x+1}$

$$\cancel{(e^x+1)} \frac{c'(x)\cancel{(e^x+1)} - \cancel{c(x)}e^x}{(e^x+1)^2} + e^x \frac{\cancel{c(x)}}{\cancel{e^x+1}} = e^x - 1$$

$$c'(x) = e^x - 1$$

$$c(x) = e^x - x + K$$

Partikularna:

$$y(0) = 0$$

$$0 = \frac{e^0 - 0 + K}{2} \rightarrow K = -1$$

$$\rightarrow \underline{\underline{y_p = \frac{e^x - x - 1}{e^x + 1}}}$$

spl. reš: $y = \frac{e^x - x + K}{e^x + 1}$

alternativno: opazimo $(e^x+1)y' + e^x y = ((e^x+1)y)'$

$$(e^x+1)y = \int (e^x - 1) dx = e^x - x + K$$

⋮

Bernoullijeva DE:

$$\boxed{y' + q(x)y = r(x)y^\alpha}$$

$\alpha=0$: LDE

$\alpha=1$: Homogena DE

splošna metoda: prevedemo na LDE: $w = y^{1-\alpha}$, $w' = (1-\alpha)y^{-\alpha}y'$

• pred tem en. delimo z y^α

→ če $\alpha > 0$, pri tem izgubimo nekatere rešitve z ničkami:

• $0 < \alpha < 1$: lahko dobimo iz ostalih z lepjenjem, dodamo $y=0$

• $\alpha \geq 1$: vsaka rešitev je 0 ali povsod 0, dodamo $y=0$

⑨ $3y' + 2y = (1+3e^x)y^4$

$$\frac{3y'}{y^4} + \frac{2}{y^3} = 1+3e^x \quad w = y^{-3} \quad w' = -3y^{-4}y'$$

$$-w' + 2w = 1+3e^x$$

$$w_h' = 2w_h$$

$$\ln \frac{w_h}{c} = 2x$$

$$w_h = ce^{2x}$$

$$\rightarrow w' = c'(x)e^{2x} + 2ce^{2x}$$

$$-e^{2x}(c'(x) + 2c(x)) + 2c(x)e^{2x} = 1+3e^x$$

$$c'(x) = -e^{-2x} - 3e^{-x}$$

$$c(x) = \frac{1}{2}e^{-2x} + 3e^{-x} + K$$

$$\rightarrow w = \left(\frac{1}{2}e^{-2x} + 3e^{-x} + K\right)e^{2x} =$$

$$= \frac{1}{2} + 3e^x + Ke^{2x} = y^{-3}$$

$$\underline{\underline{y = \left(\frac{1}{2} + 3e^x + Ke^{2x}\right)^{-\frac{1}{3}}}}$$

Dobimo še singularno reš. $y=0$.

Eksaktna NDE: $M(x,y)dx + N(x,y)dy = 0$, če $\exists F: M = \frac{\partial F}{\partial x}, N = \frac{\partial F}{\partial y}$.

Spl. reš.: $F(x,y) = C$. F je v tem primeru ohranitvena količina (1. integral enačbe).

KRITERIJ: NDE je eksaktna $\Leftrightarrow M_y = N_x$.

(12) Dokaži: NDE $(2x^3 - y)dx = (x+y)dy$ je eksaktna, izračunaj

$$\begin{aligned} M &= 2x^3 - y & M_y &= -1 \\ N &= -(x+y) & N_x &= -1 \quad \checkmark \end{aligned}$$

$$\left. \begin{aligned} \int M dx &= \frac{1}{2}x^4 - xy + A(y) \\ \int N dy &= -xy - \frac{y^2}{2} + B(x) \end{aligned} \right\} F(x,y) = \frac{1}{2}x^4 - xy - \frac{1}{2}y^2$$

$$\rightarrow \text{rešitev } \underline{\frac{1}{2}x^4 - xy - \frac{1}{2}y^2 = C}$$

Vsaka NDE je eksaktna, če jo pomnožimo s primerno funkcijo (integrirajoči množitelj / integracijski multiplikator).

$$\left(3\ln(x+y) + \frac{x}{x+y}\right)dx + \frac{x}{x+y}dy = 0$$

Namig: pomnoži s primerno potenco x .

$$\begin{aligned} M &= x^p \left(3\ln(x+y) + \frac{x}{x+y}\right) \rightarrow M_y = 3x^p \frac{1}{x+y} - \frac{x^{p+1}}{(x+y)^2} = \frac{3x^p(x+y) - x^{p+1}}{(x+y)^2} = \\ N &= \frac{x^{p+1}}{x+y} \rightarrow N_x = \frac{(p+1)x^p(x+y) - x^{p+1}}{(x+y)^2} = \frac{px^{p+1} + (p+1)x^p y}{(x+y)^2} \Bigg| = \frac{2x^{p+1} + 3x^p y}{(x+y)^2} \end{aligned}$$

$$\underline{p=2}$$

$$\int M dx = \int \left(3x^2 \ln(x+y) + \frac{x^3}{x+y}\right) dx = x^3 \ln(x+y) + A(y)$$

$$\int N dy = \int \frac{x^3}{x+y} dy = x^3 \ln(x+y) + B(x)$$

$$\rightarrow \text{spl. reš. } x^3 \ln(x+y) = C$$

